TERTIARY INFOTECH ACADEMY

LAB 03
Operate an EC2 Fleet with Systems Manager

AWS Certified SysOps Administrator Associate Training
TGS-2024051413 · Version v6.0

Scenario
Northstar Retail must remove public SSH access and make patching repeatable across tagged web instances.

Objective
Apply K3, K5, A3 and A5 by onboarding a managed node, using Session Manager and Run Command, assessing patch compliance and executing a controlled Automation runbook.

Deliverable
An EC2 managed node, SSM document, patch-compliance report, automation execution record and rollback evidence.

Services and tools
EC2 · IAM · Systems Manager · Session Manager · Run Command · Patch Manager · Automation

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Review `managed-instance.yaml`; choose the trainer-provided VPC/subnet and confirm the AMI is supported in the assigned Region.
```
aws cloudformation validate-template --template-body file://managed-instance.yaml
```

Evidence note: _____

Step 2 — Deploy the stack using a unique learner suffix and the required IAM capability.
```
aws cloudformation deploy --stack-name SysOps-Lab03-REPLACE_SUFFIX --template-file managed-instance.yaml --capabilities CAPABILITY_NAMED_IAM --parameter-overrides CourseTag=TGS-2024051413
```

Evidence note: _____

Step 3 — Obtain the instance ID from stack outputs and wait until Systems Manager reports it online.
```
aws ssm describe-instance-information --filters Key=tag:Course,Values=TGS-2024051413 --output table | tee evidence/managed-nodes.txt
```

Evidence note: _____

Step 4 — Start a Session Manager session. Confirm there is no inbound SSH rule and capture the managed shell evidence.
```
aws ssm start-session --target REPLACE_INSTANCE_ID
```

Evidence note: _____

Step 5 — Create the custom SSM document from `document-install-healthcheck.yml` and inspect its status.

```
aws ssm create-document --name SysOps-Lab03-HealthCheck --document-type Command --document-format YAML --content file://document-install-healthcheck.yml
```

Evidence note: _____

Step 6 — Run the document against the managed node with a concurrency of one and record the command ID.

```
aws ssm send-command --document-name SysOps-Lab03-HealthCheck --targets Key=tag:Course,Values=TGS-2024051413 --max-concurrency 1 --max-errors 0 | tee evidence/send-command.json
```

Evidence note: _____

Step 7 — Review Inventory and patch compliance. Explain the baseline, severity and approval rules in `CHANGE-RECORD.md`.

```
aws ssm list-compliance-items --resource-ids REPLACE_INSTANCE_ID --resource-types ManagedInstance --output table | tee evidence/compliance.txt
```

Evidence note: _____

Step 8 — Execute the read-only Automation runbook and save the output. Do not scale beyond one node until its evidence passes.

```
aws ssm start-automation-execution --document-name AWS-DescribeInstanceInformation --parameters InstanceIds=REPLACE_INSTANCE_ID | tee evidence/automation.json
```

Evidence note: _____

Step 9 — Capture Fleet Manager, compliance and command output as JPEG evidence; compare with `EXPECTED-ARCHITECTURE.jpg`.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 10 — Complete the change record with risk, owner, success metric, rollback and UTC timestamps.

```
open CHANGE-RECORD.md
```

Evidence note: _____

Step 11 — Run `validate.sh`, then remove the custom document and CloudFormation stack.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.

Acceptance criteria

- No inbound SSH from the internet
- Instance profile grants only required managed-node permissions
- Canary concurrency is one
- Patch status is explained
- Rollback and cleanup are documented

Evidence checklist

□ Managed-node inventory

□ Session Manager JPEG

□ Run Command output

□ Patch compliance export

□ Automation execution

□ completed CHANGE-RECORD.md

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.